

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTECPC402 - DATABASE MANAGEMENT SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The ability to change the conceptual schema without affecting the external schemas or application program is known _____. a. Program Data Independence b. Physical Data Independence c. Logical Data Independence d. Data Abstraction
- 2 Which level of abstraction describe What data are actually stored? a. Physical b. Logical c. External d. View
- 3 The logical design, and the snapshot of the data at a given instant in time is known as? a. Instance & Relation b. Relation & Schema c. Domain & Schema d. Schema & Instance
- 4 Either all of its operations are executed or none is called _____. a. Consistency b. Isolation c. Atomicity d. Durability
- 5 If one attribute is determinant of second, which in turn is determinant of third, then the relation cannot be: a. Well-structured b. 1NF c. 2NF d. 3NF
- 6 After groups have been established, SQL applies predicates in the _____ clause, allowing aggregate functions to be used. a. Where b. Having c. Group by d. Limit
- 7 If ____, then $A \rightarrow B$ has trivial functional dependency. a. B is a subset of A b. A is a subset of B c. A is a subset of A d. B is a superset of A
- 8 In order to build a link between two tables, which key is used? a. Primary b. Foreign c. Composite d. Unique
- 9 What is TRUE about SAVEPOINT? a. Following the completion of a transaction, it must be executed to save all the operations performed in the transaction. b. A transaction can be rolled back to its last saved state. c. A specific part of a transaction can be given a named. d. None of the above
- 10 Which command will remove the records from the table, but not affect the structure of the table? a. Remove b. Delete c. Drop d. Truncate
- 11 What is the outcome of the command `db.createCollection('myCollection')` in MongoDB? a. Creates a new document b. Deletes a collection c. Creates a new collection d. Updates a collection

12. What is a serial schedule?
 a. A schedule where transactions are interleaved.
 b. A schedule where transactions are executed one after another.
 c. A schedule with no conflicts.
 d. A schedule that is not recoverable
- Q2. Solve the following.
- A) Explain 1-tire, 2-tire, and 3-tire database architecture in detail 6
- B) Differentiate between Traditional file processing system and DBMS 6
- Q3. Solve the following.
- A) Consider the following database Emp (emp_no, emp_name, salary, department, city) Write SQL statements for following
 1. change employee name from “ Rahul” to “ Ganesh”
 2. delete the record for the employee who are from “Delhi”
 3. Out of all employees from “production” Department, list out only two employee details 6
- B) Explain the aggregate functions in SQL with their syntax and example 6
- Q4. Solve Any Two of the following.
- A) Explain 2NF, 3NF and BCNF with examples. Explain why BCNF is more strict than 3NF? 6
- B) Explain the different functional dependencies used in DBMS. 6
- C) Explain candidate key , Super key and Foreign key 6
- Q5. Solve Any Two of the following.
- A) What is Serializability in scheduling? Explain conflict Serializability in detail 6
- B) Explain lock and timestamp based concurrency control protocol. 6
- C) Explain ACID property in detail 6
- Q6. Solve Any Two of the following.
- A) State and provide a detailed explanation of CAP theorem in distributed data systems 6
- B) Explain different Types of NOSQL database. 6
- C) What are the essential CRUD operations in MongoDB? Illustrate each briefly. 6

*** End ***